

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Date	30 June 2026	Maximum Marks	4 Marks
Team ID		Project Name	Strategic Product Placement Analysis System

1. Data Flow Diagram (Simplified Flow)

The following outlines the simplified Level 0 data flow for the spatial analysis and reporting module of the Strategic Product Placement application.

- 1. Data Ingestion:** The Retail Analyst (User) uploads raw store floor plan files (PDF/PNG) and historical POS sales data (CSV) via the web interface.
- 2. Processing & Mapping:** The system's Spatial Engine parses the floor plan to generate a digital grid, mapping the POS data to specific physical aisle coordinates.
- 3. Enrichment:** Mapped data is passed to the AI Analysis Module which calculates high-traffic zones and predicts impulse-buy hotspots.
- 4. Visualization:** The enriched dataset is returned to the client and visualized as an interactive Heatmap Dashboard overlaying the original floor plan using the D3.js library.
- 5. Export:** The user requests a strategy report; the system compiles the visual data and metrics into a PDF payload and delivers it to the user's device.

2. User Stories

Below are the defined user stories for the initial MVP release of the product.

Retail Analyst (Web User)	Registration & Login	USN-1	As an analyst, I can log into the web application using my corporate email and password securely.	Successful authentication grants access to the main dashboard. Invalid credentials show an error.	High	Sprint-1
	Data Upload	USN-2	As an analyst, I can upload CSV files containing weekly product sales data and a store floor plan image.	System validates CSV structure and image format (PNG/JPG) before accepting. Shows progress bar.	High	Sprint-1
	Heatmap Visualization	USN-3	As an analyst, I can toggle a heatmap overlay on the floor plan to visually identify the highest trafficked aisles.	Heatmap renders over the floor plan within 3 seconds, using a red-to-blue color gradient based on sales volume.	High	Sprint-2
Store Manager (Mobile User)	Dashboard	USN-4	As a store manager, I can view the finalized product placement recommendations on my mobile device.	Mobile UI scales correctly. Read-only access to the finalized strategy map.	Medium	Sprint-2
	Implementation Status	USN-5	As a store manager, I can mark specific cross merchandising display setups as "Completed" on the floor plan.	Status updates in the database in real-time and reflects on the Analyst's dashboard.	Medium	Sprint-3
Administrator	User Management	USN-6	As an admin, I can create, edit, and deactivate user accounts for Analysts and Store Managers.	Admin panel lists all users. Deactivated users cannot log in.	High	Sprint-1

	System Configurations	USN-7	As an admin, I can configure the parameters of the AI enrichment module (e.g., setting the threshold for "High Traffic").	Changes reflect on all subsequently generated heatmaps across the platform.	Low	Sprint-3
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